CLASS TEST – 1

FULL MARKS = 20
MCQs – 1 * 10 = 10
SAQs – 2 * 5 = 10

Multiple Choice Questions (MCQs):

1. Who developed Python and when was it first released?
   - a) James Gosling, 1995
   - b) **Guido van Rossum, 1991**
   - c) Bjarne Stroustrup, 1983
   - d) Dennis Ritchie, 1972

   Answer: Python was created by Guido van Rossum and released in 1991.

2. Which of the following is NOT a valid Python identifier?
   - a) var_name
   - b) _variable123
   - c) **3rd_variable**
   - d) VariableX

   Answer: Variable names cannot start with a number.

3. What will be the output of the following code?
   - `x = 10`
   - `y = 3.0`

   ```
   print(type(x + y))
   ```

   - a) <class 'int'>
   - b) **<class 'float'>**
   - c) <class 'str'>
   - d) <class 'complex'>

   Answer: Adding an int and a float results in a float.

4. Which of the following statements is TRUE about Python syntax?
   - a) Python uses {} for blocks of code.
   - b) Python is case-insensitive.
   - c) **Indentation is mandatory in Python.**
   - d) Semicolons must be used at the end of every statement.

   Answer: Python uses indentation instead of {} to define code blocks.

5. What is the purpose of the input() function in Python?
   - a) To print output to the screen
   - b) To take input from the user as an integer
   - c) To take input from the user as a string

- d) **Both b and c**

Answer: input() always returns a string; it must be converted for numerical input.

6. Which of the following is NOT a Python data type?
   - a) Dictionary
   - b) Set
   - c) **Array**
   - d) Tuple

   Answer: Python has lists but does not have a built-in Array type.

7. What will be the output of the following code?

```
print(bool([]), bool(" "), bool(0), bool(1))
```

   - a) True True True False
   - b) **False True False True**
   - c) True False True False
   - d) False False False True

   Answer: Empty lists and 0 are False, non-empty strings and 1 are True.

8. What is the correct way to open and edit a Python file using IDLE?
   - a) python filename.py
   - b) open filename.py
   - c) python3 filename.py
   - d) **idle filename.py**

   Answer: To open a Python file in IDLE, use idle filename.py.

9. Which operator has the highest precedence in Python?
   - a) + (Addition)
   - b) * (Multiplication)
   - c) and (Logical AND)
   - **d) ** (Exponentiation)**

   Answer: Exponentiation has the highest precedence in Python.

10. Which statement about Python libraries is TRUE?
    - a) Python has no built-in libraries.
    - b) Libraries like NumPy and Pandas must be written from scratch.
    - c) **Python provides built-in and third-party libraries for various applications.**
    - d) Libraries must always be included in the Python standard installation.

    Answer: Python has built-in and external libraries like NumPy and Pandas.

Short Answer Questions (SAQs):

11. **Explain the difference between explicit and implicit type conversion in Python. Write a short program demonstrating both.**

    Answer:

    - **Implicit Conversion:** Python automatically converts types when needed.
    - **Explicit Conversion:** The user manually converts data types using functions like int(), float(), etc.

    Example:

```
x = 5    # int
y = 2.5  # float
result = x + y  # Implicit: int -> float
print(type(result))  # Output: <class 'float'>

num = "10"
converted = int(num)  # Explicit: string -> int
print(type(converted))  # Output: <class 'int'>
```

12. **Write a Python program that takes an integer input from the user and checks if it is even or odd using control flow statements.**

    Answer:

```
num = int(input("Enter an integer: "))
if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

13. **List and describe different types of Python tokens with examples.**

    Answer:

    - **Keywords:** Reserved words (e.g., if, for, def)
    - **Identifiers:** Variable and function names
    - **Literals:** Constants like 123, "hello", 3.14
    - **Operators:** +, -, *, /, etc.
    - **Punctuators:** :, ,, {}, ()

14. **Write a Python program that reads a string input from the user and counts the number of vowels, consonants, digits, and special characters in it.**

Answer:

```
text = input("Enter a string: ")
vowels = consonants = digits = special = 0

for char in text:
    if char.isalpha():
        if char.lower() in "aeiou":
            vowels += 1
        else:
            consonants += 1
    elif char.isdigit():
        digits += 1
    else:
        special += 1

print(f"Vowels: {vowels}, Consonants: {consonants}, Digits:
{digits}, Special: {special}")
```

15. **Python supports mutable and immutable data types. Explain the difference between them with examples. How does Python handle variable assignment for mutable and immutable objects?**

Answer:

- **Mutable:** Can be changed (e.g., list, dict, set).
- **Immutable:** Cannot be changed (e.g., int, float, str, tuple).

Example:

```
# Mutable
lst = [1, 2, 3]
lst[0] = 100  # Allowed

# Immutable
s = "hello"
s[0] = "H"  # Error: Strings are immutable
```

Variable Assignment:

- Immutable objects create a new memory reference when changed.
- Mutable objects modify the same reference in memory.